Run sjSDM Tuning Candidates

Description

Fits every deterministic regularization candidate independently on each cross-validation training partition and scores held-out probabilities.

Usage

```
run_sjsdm_tuning_candidates(
  data_assignments = NULL,
  data_candidates = NULL,
  prepare_fold_function = NULL,
  fit_function = NULL,
  predict_function = NULL,
  score_function = score_sjsdm_tuning_predictions,
  seed = 900723L,
  epsilon = 1e-06,
  retain_prediction_cache = FALSE
)
```

Arguments

`data_assignments`	Location-level assignment table with 'repeat_id', 'fold_id', 'location_id', 'n_samples', and 'row_indices'. Optional 'cv_strategy' is propagated.
`data_candidates`	Candidate table returned by [build_sjsdm_regularization_candidates()].
`prepare_fold_function`	Injectable function called with 'train_indices', 'test_indices', 'repeat_id', and 'fold_id'. It must return 'data_train_input', 'data_test_input', and matrix 'data_test_observed'.
`fit_function`	Injectable function called with 'data_train_input', one-row 'candidate', and deterministic integer 'seed'.

`predict_function`	Injectable function called with fitted 'object' and 'data_test_input'. It must return a probability matrix aligned to 'data_test_observed'.
`score_function`	Injectable function called with the fitted object, test input, aligned observations and probabilities, epsilon, and, when supported, a deterministic 'score_seed'. Defaults to marginal binary prediction scoring; production sjSDM tuning supplies the joint-likelihood scorer.
`seed`	Single non-negative integer used to derive stable fit and score seeds. Defaults to '900723L'.
`epsilon`	Probability clipping tolerance used for held-out log loss. Defaults to '1e-6'.
`retain_prediction_cache`	Logical. When true, retain compact fold metadata and candidate probability matrices so selected OOF artifacts can be assembled without refitting.

Details

Fold preparation is run once per repeat and fold. Preparation, fit, prediction, and scoring errors are retained as structured status rows so a failed candidate does not abort the remaining tuning grid.

Value

Compact tibble with one row per repeat, fold, and candidate. It contains candidate parameters, fold counts, total and per-response held-out negative log likelihood, macro AUC, fit status, error text, CV strategy, and regularization source. Fitted objects and prediction matrices are omitted by default. With 'retain_prediction_cache = TRUE', a named list contains the unchanged tuning table and one compact cache entry per fold.